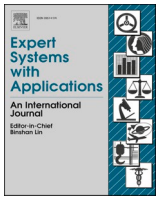
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Cooperative task allocation with simultaneous arrival and resource constraint for multi-UAV using a genetic algorithm

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ABSTRACT

In recent years, the application of multi-UAV cooperation systems has expanded across various domains. Enhancing the coordination performance of multi-UAV systems can be achieved through task allocation methods, typically relying on a hierarchical structure. This paper proposes a novel approach using a modified genetic algorithm (GA) to address the integrated task allocation and path planning problems for multi-UAV attacking multi-target. To create a more realistic mission scenario, multiple constraints, such as resource requirement and simultaneous target arrival, are considered. The modified GA incorporates tailored crossover and mutation operators that ensure compliance with the aforementioned constraints. Furthermore, an unlocking strategy is devised to prevent the occurrence of a chromosome deadlock condition, in which several UAVs become stuck in an infinite waiting state. Through simulation results, the modified GA is demonstrated to effectively delivers feasible solutions to the coupled task allocation and path planning problems, preserving the integrated nature of the optimization process. Monte Carlo simulations are conducted to highlight the superiority of the proposed method in comparison to conventional approaches.

1. Introduction

Multi-UAV coordination can overcome the shortcomings of single UAV, which plays a vital role in both civilian and military aspects, such as data collection (Wei et al., 2022), cooperative search (Fei et al., 2022) et al. Two primary challenges in coordinating multiple UAVs are task allocation and path planning. The autonomous mission, whether involving homogeneous or heterogeneous UAVs, necessitates the consideration of various constraints (Xiong et al., 2023).

To solve the multi-UAV task allocation problem, the existing methods are usually classified as deterministic, heuristic (Mohamadi, Kara, & Lagha, 2022) and market-based methods (Hooshangi, Alesheikh, Panahi, & Lee, 2021). After acquiring each UAV's task sequences through the task allocation process, the path planning process is initiated to generate flyable and safe trajectories for each UAV, with the objective of minimizing or maximizing a predefined function, such as finding the shortest flight path. Several basic constraints, including curvature bounds, collision avoidance with other UAVs, and no-fly zones (NFZs), are considered in the path planning problem.

Traditionally, task allocation and the path planning problems have been treated as two independent sub-problems in the above studies. However, it is essential to recognize that the objective function of the task allocation problem is dependent on the path length, while the accurate path length can only be determined based on the allocation strategy. As a result, the task allocation and path planning problems are inherently coupled, influencing each other's solutions and outcomes (Zhen, Chen, Wen, & Han, 2020).

Some studies have addressed the task allocation and path planning problem for multi-UAV systems by focusing on integrated solutions (Liu, Wei, Wang, Li, & Wang, 2021). These approaches successfully obtain task sequences and Dubins paths for the UAVs. However, most of these studies have overlooked the real-life scenario by assuming fixed allocation relationships between UAVs and targets. In real-life scenario, different targets may have varying and flexible resource requirements, often expressed as multi-dimensional vectors. In such cases, UAV coalitions may need to be assigned to attack a target based on its resource

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requirement, which could lead to assignment deadlocks due to overlapping coalitions (Oh, Kim, Ahn, & Choi, 2017).

To the above problems, this paper focuses on the integrated solution for the coupled task allocation and path planning problems for multiple UAVs are tasked with attacking multiple targets. In contrast to prior works, this paper introduces an enhancement by considering feasible resource requirements for the targets in this integrated problem. Incorporating resource requirements can result in coalition formation and assignment deadlock issues, which necessitate innovative solutions. To tackle these challenges, a modified genetic algorithm (GA) is proposed to solve the integrated optimization problem efficiently. Furthermore, to model UAV flight paths in the coupled task allocation and path planning problems, Pythagorean Hodograph (PH) curves are adopted. The use of pH curves ensures curvature continuity in each UAV's flight path, the length, curvature and line shape of pH curve can be altered by modifying the modulus of the direction vectors at either the start or end points, enhancing the overall efficiency and safety of the mission. The main contributions of the work are listed as follows:

- A novel heterogeneous multi-UAV task allocation model is presented, which incorporates considerations such as resource requirements, simultaneous target arrival, and the utilization of pH curves, collectively creating a more realistic mission scenario.
- A modified GA with multi-type genes is proposed, enabling it to simultaneously address the coupled task allocation and path planning problems. In this approach, an unlock mechanism is developed that ensures the generation of deadlock-free chromosomes based on situation information. This enhancement empowers the GA to handle intricate scenarios with overlapping coalitions and delivers more robust solutions. The integration of these constraints and the multi-type gene approach results in more effective and practical solutions for real-world multi-UAV missions.

The remainder of this paper is organized as follows: Section 2 presents the related works on multi-UAV task allocation. In Section 3, the problem statement and model formulation based on directed graph are presented. Then, a modified GA is proposed to solve the integrated optimization problem in Section 4. Simulation and experimental results are given in Section 5. The conclusions are shown in Section 6.

2. Related works

Task allocation is the process of assigning a number of UAVs to a set of targets or tasks under a variety of constraints and determining each UAV's task sequence, which can minimize or maximize a predefined objective function (Coutinho, Battarra, & Fliege, 2018). These constraints encompass target-related (Pilloni, Ning, & Atzori, 2021) factors, such as the priority of targets, resource requirement, and simultaneous arrival constraints. Additionally, UAV-related constraints, such as UAV kinematics, fuel limitations, and collision avoidance among UAVs, play a crucial role. Furthermore, environment-related constraints, such as collision avoidance with obstacles and no-fly zones, must also be taken into account (Wood, Khoo, Michael, Manzie, & Shames, 2020). As the number of UAVs and targets in the mission increases, the complexity of the coupled task allocation and path planning problems escalates significantly (Wang & Zhang, 2022).

The multi-UAV task allocation problem is usually transformed into a classical combinatorial optimization problem, some common formulations include the multiple traveling salesman problem (MTSP) (Zhan & Zeng, 2019; Cai, Zhang, & Zheng, 2017), vehicle routing problem (VRP) (Rojas Viloria, Solano-Charris, Muñoz-Villamizar, & Montoya-Torres, 2021), mixed-integer linear programming (MILP) (Murray & Raj, 2020), the generalized assignment problem (GAP) (Amorim, Alves, & de Freitas, 2020), game theory (Chen et al., 2020; Li, Li, Vo, & Kowalczyk, 2022), Markov process (Zhao, Ye, Pei, Liang, & Niyato, 2022), and cooperative multiple task assignment problem (CMTAP). For example, a game formulation is used to model the task allocation problem considering the budget constraint (Li et al., 2022). In the multi-UAV mobile edge computing, a Markov decision process (MDP) is used to formulate the task offloading optimization problem (Zhao et al., 2022). In practice, the complexity of the mission being performed often leads to the modeling of task allocation as a variation of the above problems to suit specific scenarios. To address the limitations of UAV capacity and task time windows, two crucial constraints are introduced into the VRP framework, resulting in two variant problems: the capacity-VRP (CVRP) (Hintsch & Irnich, 2018) and VRP with time windows (VRPTW) (Stecz & Gromada, 2020), respectively.

The existing task allocation methods are usually classified as deterministic, heuristic and market-based methods. Typical deterministic methods such as the branch and bound algorithm, depth/breadth-first search, and dynamic programming, seek to find the optimal solution of the problem based on some assumptions. In contract, the heuristic methods attempt to emulate nature phenomena, often reflecting the intelligence observed in groups or swarms (Tang, Duan, & Lao, 2023). These algorithms, including ant colony optimization (Alhaqbani, Kurdi, & Youcef-Toumi, 2020) (ACO), particle swarm optimization (Tang et al., 2021) (PSO), simulated annealing (SA) and genetic algorithm (Sun et al., 2022) (GA), are widely utilized to solve task allocation problems across various mission scenarios. Market-based task allocation methods (Hooshangi et al., 2021) have evolved from the contract net protocol, offering agents a simulated economic market for trading tasks and resources.

Because the optimization objective of the task allocation problem often includes path cost, some research studies have approached the coupled task allocation and path planning problem by treating it as an integrated optimization problem. Typically, stochastic algorithms are employed to solve this integrated optimization problem, aiming to obtain both the task sequence and feasible paths for each UAV simultaneously. Tree representation is adopted to model the CMTAP (Shima, Rasmussen, Sparks, & Passino, 2006) scenario, wherein the objective function seeks to minimize the total path length of all UAVs using the Dubins path. To handle the CMTAP alongside the Dubins path planning problem, heading angle discretization and directed graphs are utilized (Edison & Shima, 2011), two genetic-based algorithms are proposed to deal with two situations where the UAVs are homogeneous or heterogeneous, respectively. The chromosomes in these genetic algorithms serve to represent both the task allocation and path planning information concurrently. Furthermore, some studies have expanded the integrated optimization of the coupled CMTAP and Dubins path planning problem by considering additional constraints (Deng, Yu, & Wang, 2013; Deng, Yu, & Mei, 2014; Wu, Wang, & Cui, 2018; Jia, Yu, Ai, Xu, & Yang, 2018). To model the limited resources of UAVs and address deadlock chromosomes in the CMTAP, a mirror representation is introduced, analyzing the task precedence graph (TPG) (Deng et al., 2014). Moreover, the centralized genetic algorithm is extended to a distributed form, offering a more scalable approach to the integrated optimization problem (Wu et al., 2018). To depict a more realistic mission scenario, stochastic velocities of UAVs and time windows for different tasks are taken into account (Jia et al., 2018).

3. Problem statement and model formulation

In this paper, a scenario that a team of heterogeneous fixed-wing UAVs is assigned to attack a set of known ground targets. The integrated task allocation and path planning problems are formulated as a combinatorial optimization problem. Different from the traditional task allocation problem, this paper focuses on the integrated optimization strategy for the coupled task allocation and planning problems while considering the coalition formation caused by the resource requirement of targets.

In addition, some assumptions are made to simplify the problem: (1) the fuel restriction constraint of UAVs is considered equivalent to their

maximum flight distance; (2) the UAVs can communicate normally without any communication interference or interruptions; (3) UAVs will not be destroyed during the mission.

3.1. UAVs

Assume that $U = \{U_i | i = 1, 2, ..., N_u\}$ is the set of N_u heterogeneous UAVs. The heterogeneity of the UAVs is reflected in the diverse mission loads they carry. For Instance, reconnaissance-oriented UAVs are equipped with a variety of sensors specifically designed for reconnaissance purposes. Conversely, attack mission-oriented UAVs are equipped with various types of missiles suitable for carrying out attack missions. Each UAV can carry limited number of onboard mission loads which can be described as a n-dimensional row vector $\boldsymbol{R}^{U_i}$ as follows:

$$\boldsymbol{R}^{U_i} = \left(R_1^{U_i}, R_2^{U_i}, ..., R_n^{U_i}\right) \tag{1}$$

where $R_h^{U_i} (h = 1, ..., n)$ denotes the amount of type-h resource carried by UAV U_i, n represents the types of mission loads carried by the UAV U_i.

Assuming that each UAV can fly at a fixed height and the spatial configuration of UAV U_i can be denoted as $q_i = [x_i, y_i, \chi_i]^T$, with the following kinematic constraint:

$$\dot{q}_i = \begin{bmatrix} \dot{x}_i \\ \dot{y}_i \\ \dot{\chi}_i \end{bmatrix} = \begin{bmatrix} \cos\chi_i & 0 \\ \sin\chi_i & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_i \\ \omega_i \end{bmatrix} \tag{2}$$

where (x_i, y_i) denotes the horizontal coordinates of UAV U_i in a Cartesian inertial reference frame, v_i is the constant velocity of UAV U_i, χ_i is the heading angle and ω_i is the turn rate. ω_i also severs as the control input of UAV U_i as follows:

$$\omega_i = \frac{c \cdot v_i}{\rho_{\min}^i} \tag{3}$$

where $\rho_{\min}^i$ denotes the minimum turning radius of UAV U_i, $c, (|c| \le 1)$ represents the steering command.

3.2. Targets

Suppose that $T = \{T_j | j = 1, ..., N_t\}$ is the set of N_t stationary ground targets which is known to UAVs at the initial time. In a realistic battlefield, for each target, multiple types and numbers of ammunition may be required to destroy it. To describe the resources required to destroy the target, the resource requirement of target T_j can be expressed as follows:

$$\boldsymbol{R}^{T_j} = \left(R_1^{T_j}, R_2^{T_j}, ..., R_m^{T_j}\right), j \in \{1, ..., N_t\} \tag{4}$$

where $R_q^{T_j} (q = 1, ..., m)$ means the number of type-q resources needed to destroy the target T_j, m represents the types of resources needed by the target T_j.

Let $Value = (Va_1, ..., Va_{N_t})$ be the initial value of the target set. The score function of target T_j can be expressed as follows:

$$s_j(t_j) = Va_j \cdot e^{-\lambda_j\left(t_j - t_j^0\right)} \tag{5}$$

where $s_j(t_j)$ means the score when the target T_j is performed at time t_j. $\lambda_j > 0$ denotes the time-discounting factor for the target T_j. t_j^0 is time when target T_j is generated.

3.3. Directed graph representation

In the context of coupled task allocation and path planning problems, the integrated model should not only capture the allocation relationships between UAVs and targets but also define feasible paths for the UAVs. To accomplish this, binary decision variables are employed to represent the task allocation strategy between UAVs and targets, while heading angles at the starting and ending points determine the UAVs' paths. However, the continuous changes in heading angles during UAV flight introduce numerous possibilities for the angles at the starting and ending points, leading to computational complexities in the optimization process. To address this challenge and reduce calculation complexity, the heading angle discretization method were adopted. Based on the directed graph representation, the integrated task allocation and path planning problems are formulated.

The discretized heading angle set H can be expressed as:

$$H = \{\varphi_i | \varphi_i = 2\pi i / N_\varphi, i = 0, 1, ..., N_\varphi - 1\} \tag{6}$$

where N_φ, $(N_\varphi \ge 1 \cup N_\varphi \in Z)$ denotes the degree of the discretization. For example, when $N_\varphi = 6$, the heading angle set is $H = \{0, \pi/3, 2\pi/3, \pi, 4\pi/3, 5\pi/3\}$.

According to the heading angle discretization method, each target T_j, $j \in \{1, ..., N_t\}$ can be represented as a set of vertices in a directed graph as follows:

$$\boldsymbol{V}_{T_j} = \left\{\left(T_j, \varphi_1\right), \left(T_j, \varphi_2\right), ...\left(T_j, \varphi_{N_\varphi}\right)\right\}, j \in \{1, ..., N_t\} \tag{7}$$

In the graph, the set of vertices of the target T_j is denoted by $\boldsymbol{V}_{T_j}$. Each node is identified by the position and heading angle of T_j, $\{\varphi_1, \varphi_1, ..., \varphi_{N_\varphi}\}$ is the set of discretized heading angles. The vertices in the directed graph of the coupled task allocation and path planning problem can be denoted as:

$$\begin{cases} \boldsymbol{V_T} = \left\{\boldsymbol{V}_{T_1}, \boldsymbol{V}_{T_2}, ..., \boldsymbol{V}_{T_{N_t}}\right\} \\ \boldsymbol{V_U} = \left\{\boldsymbol{V}_{U_1}, \boldsymbol{V}_{U_2}, ..., \boldsymbol{V}_{U_{N_t}}\right\} = \{(U_1, \varphi_1), (U_2, \varphi_2), ..., (U_{N_u}, \varphi_{N_u})\} \\ \boldsymbol{V_{all}} = \boldsymbol{V_T} \cup \boldsymbol{V_U} \\ \|\boldsymbol{V_T}\| = N_t \cdot N_\varphi \\ \|\boldsymbol{V_U}\| = N_u \\ i \in \{1, ..., N_t\}, j \in \{1, ..., N_\varphi\} \end{cases} \tag{8}$$

where $\boldsymbol{V}_T$ and $\boldsymbol{V}_U$ are the sets of vertices representing targets and UAVs, respectively. V_{all} is the set of all vertices in the directed graph. The directed edge set of the directed graph can be represented as:

$$\boldsymbol{E} = \left\{\left(v_i, v_j\right) \middle| v_i \in \boldsymbol{V_{all}}, v_j \in \boldsymbol{V_T}\right\} \tag{9}$$

Here below is an example denoting the directed graph and task planning (task allocation and path planning) results of four UAVs attacking two targets. The initial heading angle of the UAVs are denoted as $\phi_i, (i = 1, 2, 3, 4)$. Two targets are required to be attacked by these UAVs, and the degree of discretization of heading angle $N_\varphi = 3$, therefore the possible heading angle set for each target is $\{0, 2\pi/3, 4\pi/3\}$.

Fig. 1(a) shows an example of a directed graph $G(V_{all}, E)$ represents UAV U_1 and U_3 cooperative attack the target T_1 with heading angle 0° and 240°, respectively. UAV U_2 and U_4 cooperatively attack the target T_2 with heading angle 0° and 120°, respectively. The task planning results in a realistic environment is shown in Fig. 1(b). The Pythagorean Hodograph (PH) curve is used as the path for these UAVs, because the curvature of pH curve is continuous.

3.4. Combinatorial optimization problem

In the context of a multi-UAV cooperative attacking mission, the targets and UAVs are represented as in the directed graph $G(V_{all}, E)$, while the paths of the UAVs are depicted as edges connecting these vertices. The task allocation and path planning problems for multi-UAV missions attacking multiple targets can be formulated as an integrated combinatorial optimization problem. For this optimization problem, adopt two criteria to quantify the objective function: the cumulative flight distance of all UAVs and the maximum length of flight distance covered by any UAV. Hence, the objective function (denoted as J) to be

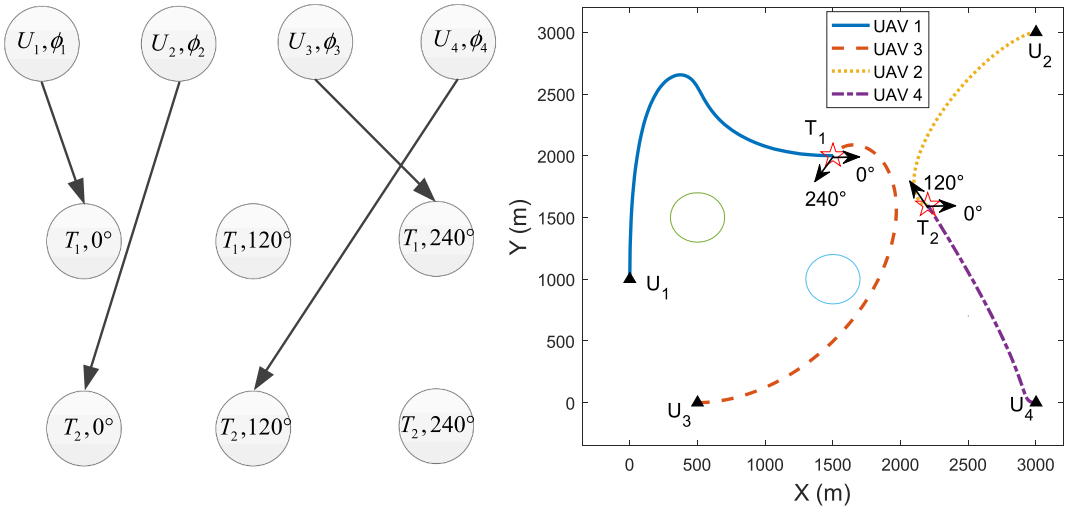


Fig. 1. (a) a directed graph of the task planning problem; (b) the task planning results.

minimized can be defined as follows:

$$J=\sum_{h=1}^{N_u}\sum_{i=1}^{\|V_{all}\|}\sum_{j=1}^{\|V_T\|}x_{(v_i,v_j)}^{U_h}w_{(v_i,v_j)}^{U_h}+\alpha\cdot\left(\max_{U_h}\sum_{i=1}^{\|V_{all}\|}\sum_{j=1}^{\|V_T\|}x_{(v_i,v_j)}^{U_h}w_{(v_i,v_j)}^{U_h}\right) \tag{10}$$

where $w_{(v_i,v_j)}^{U_h}$ is the cost of UAV U_h moving from the vertex v_i to v_j, which equals to the path length of the PH curve between these two nodes. α is a user-defined coefficient. Increasing the value of α can reduce the mission completion time, reducing the value of α can decline the cumulative flight distance of all UAVs. $x_{(v_i,v_j)}^{U_h}$ is a binary decision variable:

$$x_{(v_i,v_j)}^{U_h}\in\{0,1\} \tag{11}$$

where $x_{(v_i,v_j)}^{U_h}=1$ denotes that the UAV U_h will perform the target represented in vertex v_j following the edge $e=(v_i,v_j)$.

In the heterogeneous multi-UAV cooperative attacking mission, the following constraints need to be considered:

(1) Limited resources of UAVs

Different UAVs are equipped with varying resources in terms of both quantity and size. Each UAV is capable of targeting multiple objectives based on its task sequence. It is crucial to ensure that the cumulative resource consumption throughout a UAV's task sequence does not surpass the resources carried by that specific UAV, which can be expressed as:

$$\sum_{i=1}^{\|V_{all}\|}\sum_{j=1}^{\|V_T\|}R_{(v_i,v_j)}^{U_h}\le R^{U_h},\ \forall h\in\{1,\ldots,N_u\} \tag{12}$$

where $R_{(v_i,v_j)}^{U_h}$ denotes the resource consumed of UAV U_h following the edge $e=(v_i,v_j)$ to attack the assigned target.

(2) Resource requirement of targets

To attack a target successfully, the resource requirement of targets should be satisfied. So, a UAV coalition maybe assigned to attack a same target. The total resources of the UAV coalition assigned to attack the target T_j should satisfy:

$$\sum_{h=1}^{N_u}\sum_{i=1}^{\|V_{all}\|}\sum_{l=0}^{N_\phi-1}R_{(v_i,(T_j,\phi_l))}^{U_h}\ge R^{T_j},\quad \forall T_j\in\{1,\ldots,N_t\} \tag{13}$$

where $R_{(v_i,(T_j,\phi_l))}^{U_h}$ means the resource consumed of UAV U_h attacking the assigned target T_j with ending angle ϕ_l.

(3) Limited fuel of UAVs

In this paper, the UAVs fly at a constant speed, the fuel constraint of UAV can be approximately equivalent to the maximum flying distance of UAV:

$$\sum_{i=1}^{\|V_{all}\|}\sum_{j=1}^{\|V_T\|}x_{(v_i,v_j)}^{U_h}w_{(v_i,v_j)}^{U_h}\le L_{\max}^{U_h},\quad h\in\{1,\ldots,N_u\} \tag{14}$$

where $L_{\max}^{U_h}$ means the maximum flying distance of UAV U_h.

(4) Maximum number of UAVs that can be allocated to a single target

Considering the realistic mission environment, it is generally required to allocate no more than N_j^T UAVs to attack the target T_j:

$$\sum_{h=1}^{N_u}\sum_{i=1}^{\|V_{all}\|}\sum_{l=0}^{N_\phi-1}x_{(v_i,(T,\phi_l))}^{U_h}\le N_j^T,\quad \forall T_j\in\{1,\ldots,N_t\} \tag{15}$$

(5) The deadlock constraint of task sequence of UAVs

Given the consideration of resource requirements and simultaneous attack, a particular situation known as a 'deadlock' may arise. In this context, a deadlock refers to condition in which several UAVs find themselves in a state of mutual waiting, leading to an infinite waiting scenario. Here is an example to illustrate the deadlock condition. There are three UAVs in the environment. The task sequences of the UAVs are $A^{(1)}=[1,2,3,4]$,$A^2=[1,4]$ and $A^3=[1,4,3]$, which means that the UAV U_1 will attack the target $T_1,T_2.T_3,T_4$ in order, U_2 will attack T_1 and then T_4, UAV U_3 attack T_1,T_4,T_3 in order. In addition, the UAVs assigned to attack the same target should arrive the target simultaneously. It can be seen from the task sequences of theses UAVs that, after the UAV U_1 attack target T_2, U_1 waits for U_3 to simultaneous arrival T_3. However, for UAV U_3, it waits for U_1 to simultaneous attack T_4. So, the two UAVs U_1 and U_3 fall into endless waiting and cannot attack the target T_3 and T_4.

Here is the mathematical description of the deadlock. The task sequences of multi-UAV without deadlock are equivalent to the directed graph $G(T,\varepsilon(A))$ as a directed acyclic graph (DAG) (Al-Hawri, Correia, & Barradas, 2020; Gissibl & Klüppelberg, 2018):

$$\begin{cases} G(T,\varepsilon(\mathrm{A}))=DAG \\ T\{1,\ldots,N_T\} \\ \varepsilon(\mathrm{A})=\left\{\left(\mathrm{A}^{(i)}(k),\mathrm{A}^{(i)}(k+1)\right)\middle|i\in\{1,\ldots,N_u\}\right\}\end{cases} \tag{16}$$

To detect the above deadlock example based on this principle, the directed graph $G(T,\varepsilon(A))$ of the task sequences of UAVs in the above example is shown in Fig. 2.

It can be seen from the above directed graph $G(T,\varepsilon(A))$ that, there is a directed ring between nodes T_3 and T_4. So, the task allocation strategy will cause deadlock condition.

In summary, the problem of multi-UAV cooperative attack mission can be formulated as a combinatorial problem. The objective function is

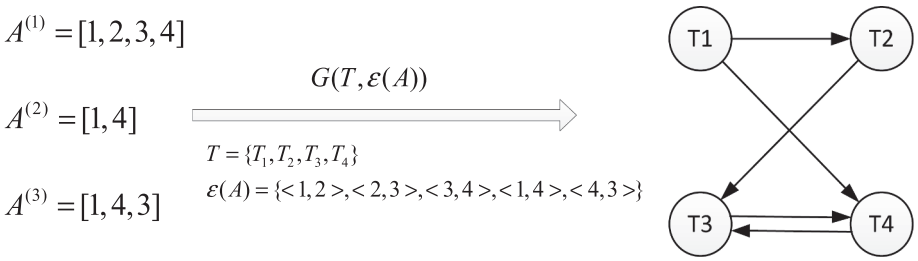


Fig. 2. The task sequences of UAVs and the corresponding directed graph $G(T, \varepsilon(A))$.

represented by Equation (10), subject to the constraints outlined in Equations (12) ~ (16).

the multi-UAV cooperative attacking mission can be modeled as a combinatorial optimization problem, whose objective function is shown in Equation (10) with constraints Equations (12)-(16).

4. Modified GA for multi-UAV task allocation

The integrated optimization model of multi-UAV cooperative attack mission exhibits the following key characteristics: 1) The model encompasses the allocation strategy between UAVs and targets, as well as detailed flight path information for each UAV; 2) coordination requirements among UAVs are primarily centered on each UAV's necessity to collaborate with others for achieving simultaneous attacks and avoiding potential deadlock, all while executing their individual task sequences. 3) Consider the resource requirements of targets, the resource allocation is coupled with the task allocation and path planning process (Zhou et al., 2019; Xing, Liu, Xu, & Nallanathan, 2019).

In view of the characteristics of the problems considered in this paper, a modified genetic algorithm (GA) with multi-type genes is designed in this section. Firstly, in terms of encoding, the information of task allocation and path planning are integrated encoded, and a method to determine the execution sequence of targets is proposed based on the battlefield situation, so that the encoded chromosomes can avoid the deadlock condition. After that, the chromosome crossover and mutation operators are designed which can reflect the characteristics of the problem. Finally, for the resource requirement of targets, the resource feasibility of chromosome is checked, the chromosomes that do not satisfy the resource requirement constraint are modified to obtain viable chromosomes. When implementing the modified GA to solve the integrated optimization problem, the flow chart is shown in Fig. 3. From the flow chart, the objective function of the integrated optimization problem depends on the path length of each UAV. As a result, the fitness evaluation of each chromosome is closely tied to the path generation process during the GA's execution

4.1. Encoding

The most critical part in the modified GA is the encoding of chromosome. For the integrated optimization problem, the encoding form of chromosome is shown in Fig. 4.

Each chromosome can be seen as a matrix composed of three row vectors, each column in the chromosome is seen as a gene denotes a UAV with a heading angle to attack a target. The maximum number of UAVs that can be allocated to a target N_j^T is seen as the number of genes to represent the allocation strategy of the target T_j. So, the total genes of a

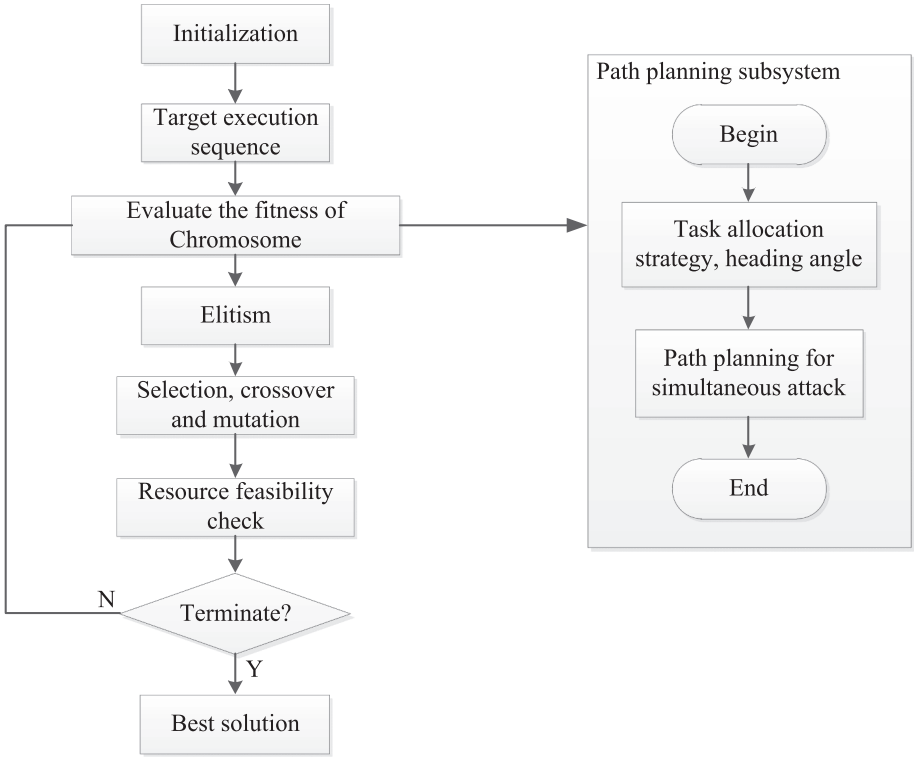


Fig. 3. Flow chart of the modified GA for the integrated optimization problem.

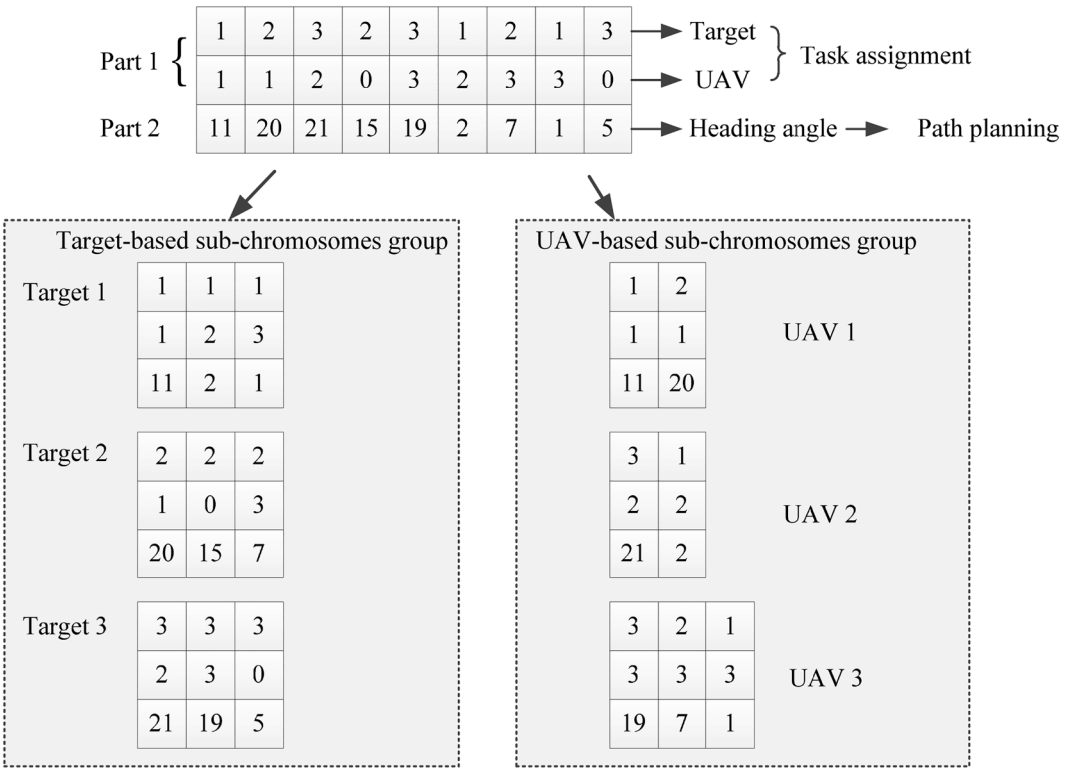


Fig. 4. An example of the chromosome encoding in the modified GA ($N_1 = 36$).

chromosome are $N_1^T + ..., + N_{N_t}^T$. Each gene is composed of two parts: the first part denotes the allocation strategy between UAVs and targets. The second part stands for the heading angle information of UAV to reach the target, which can be selected from the set $\phi \in \{0, 1, ..., N_\varphi - 1\}$ and the actual heading angle is $\varphi = 2\pi\phi/N_\varphi$.

Let $N_j^T = 3, j \in \{1, ..., N_t\}$, which means that for any target, a maximum number of three UAVs can be allocated to attack the target. So, 0 element may appear in the second row of a gene, which means that the gene has no practical effect. The example chromosome represents that, UAV U_1, U_2, U_3 are allocated to simultaneous attack the target T_1 with heading angle $110°, 20°, 10°$, respectively. U_1 and U_3 are required to attack T_2 with heading angle $200°$ and $70°$, respectively. U_2 and U_3 are allocated to attack T_3.

Furthermore, a chromosome can be transformed into two sub-chromosomes based on targets and UAVs respectively (illustrated in the bottom half of Fig. 4). The genes with the same target number in the chromosome can be extracted to form a target-based sub-chromosomes group, which contains N_t target-based sub-chromosomes. Each target-based sub-chromosome denotes the UAVs allocated to attack the target. Since the UAVs within the same target-based sub-chromosome are required to attack the target simultaneously, the gene sequence within the target-based sub-chromosome can be adjust arbitrarily. As a result, when a chromosome is encoded, the corresponding target-based sub-chromosomes group is unique.

By extracting the genes with the same UAV number from the chromosome, it is possible to transform one chromosome into a UAV-based sub-chromosomes group. This group consists of no more than N_u UAV-based sub-chromosomes, with each sub-chromosome representing the allocation strategy of a specific UAV. The number of genes in the UAV-based sub-chromosome is equal to the number of the targets that the corresponding UAV needs to attack.

4.2. Initialization and fitness calculation

The chromosome initialization operator obtains the initial population for the optimization problem, which be divided into the following steps:

Step 1. For each target, initialize three genes. Each gene is represented as a column consisting of three elements. Populate the first line of the three genes corresponding to the target with the target number.

Step 2. For any gene of a certain target, randomly select an element belonging to the set $\{0, 1, ..., N_u\}$ and put it in the second line of the gene.

Step 3. For each gene, randomly select a target arrival heading angle from the set $\{0, 1, ..., N_\varphi - 1\}$ in the third line of the gene.

After the initialization of the chromosomes, the fitness calculation process is required to evaluate each chromosome. The fitness of a chromosome reflects the quality of solution represented by the chromosome. The fitness of chromosome is inversely proportional to the objective function as follows:

$$f = 1/J \tag{17}$$

where J denotes the objective function of the integrated optimization problem as shown in Equation (10). A chromosome's fitness is based on the summation of edge weight. Each edge represents the length path of pH curve between two vertices with initial and final heading angles. To a certain target, a UAV coalition maybe allocated to attack the target, which requires the UAV coalition to arrive the target simultaneously. To generate the paths for multi-UAV arriving a certain target, a simultaneous arrival path planning method is used according to the previous work, which is described in detail in reference (Yan, Zhu, Zhou, & Tang, 2019). The simultaneous arrival path planning method use the distributed cooperative particle swarm optimization to generate the PH curves with equal length for UAVs and avoid collision with no-fly-zones by adjusting the particles' fitness. After the fitness calculation process of all the chromosomes in a generation, parental chromosomes are selected to generate the offspring chromosomes based on the roulette wheel method.

4.3. Genetic operators

(1) Crossover operators

The crossover operator is a procedure in which parental chromosomes produce offspring chromosomes by exchanging genetic information at corresponding gene positions. Two parental chromosomes can generate two child chromosomes through crossover operation. In general, since the offspring chromosomes inherit the superior genes from their parental counterparts, they typically exhibit improved fitness compared to the parental chromosomes.

A single site crossover method is designed and an example is illustrated in Fig. 5. The mission scenario involves four UAVs attacking three targets. After the initialization process of chromosomes, two parent chromosomes A and B are probabilistically chosen from the current generation in step 1. In step 2, firstly, the genes of Parent A are disrupted and rearranged so that the genes with different target numbers have the equivalent probability to perform crossover operations. Secondly, the crossover site s is randomly selected from $[1, N_c]$ (in Fig. 5, $N_c = 9, s = 7$), and the previous $s-1$ genes are copied to the generated offspring without any change, the remaining $N_c - s$ genes of the offspring are generated by crossover operator. In step 3, in order to avoid violating the maximum number of UAVs allocated to a single target, see Equation (15), all the targets of Parent A are copied to the offspring. For the current crossover site 7 to be executed, since the allocation strategy of target T_3 is UAV U_1 and U_3 in Parent A, these two UAVs cannot appear in the site 7 of the offspring chromosome. The attack complement set of Parent B which denotes the genes with the ability to attack except U_1 and U_3 are extracted, as shown in the right part of step 3. In step 4, an attack gene from the attack complement set of Parent B, is randomly selected and copied to the child's genes. The first offspring is obtained by repeating the above process for each of the reaming genes $i \in [s, N_c]$. Analogously, the creation of the second child only needs to exchange the positions of the Parent A and B.

(2) Mutation operators

Mutation is a critical mechanism within the genetic algorithm, responsible for introducing new genetic material into the population by altering one or more genes in a chromosome. This process plays a pivotal role in enhancing population diversity and preventing chromosomes from getting trapped in local optima. In this paper, two distinct mutation techniques are employed: (1) heading angle mutation; (2) mutation in UAV allocation strategy.

Heading angle mutation is performed by randomly select a new heading angle $\phi_{new} \in H \backslash \{\phi_{old}\}$ to replace the old one. Although the heading angle mutation only changes one heading angle of a chromosome, it can not only change the flight path of the UAV represented by the gene, but also change the flight path of other UAVs in the same coalition, and change the subsequent flight path of these UAVs.

In the second way of mutation, assume that the target of the mutant gene is T_j, a new UAV $U_{new} \in U \backslash \{U_{old}\}$ is randomly selected to replace the old one. Here, U_{old} is shown as follows:

$$U_{old} = \left\{ u \left| \sum_{i=1}^{\|V_{all}\|} \sum_{l=0}^{N_\phi - 1} x^u_{\left(v_i\left(T_j, \phi_l\right)\right)} = 1 \right. \right\} \tag{18}$$

To express the second way of mutation more clearly, an example is used to illustrate this mutation way in Fig. 6.

4.4. Unlock the deadlock chromosome

As discussed in Section 3.4, the presence of a deadlock condition in the chromosome evolution process is a significant concern. Deadlock arises when multiple UAVs find themselves in a perpetual cycle of waiting of each other, resulting in an indefinite waiting state and hindering the algorithm's effective progress.

In traditional studies (Deng et al., 2014; Jia et al., 2018), the task sequence of a UAV is determined by the genes sequence in the corresponding UAV-based sub-chromosome. The commonly used method of unlocking the deadlock chromosome is to adjust the sequence of the genes in the UAV-based sub-chromosome, which can change the target execution sequence of UAVs. However, when dealing with a large number of UAVs and targets, the deadlock condition can become more complex. In such cases, a greater number of UAVs may be involved in the deadlock, leading to more intricate deadlock situations that require the movement of a larger number of genes to unlock the deadlock chromosome.

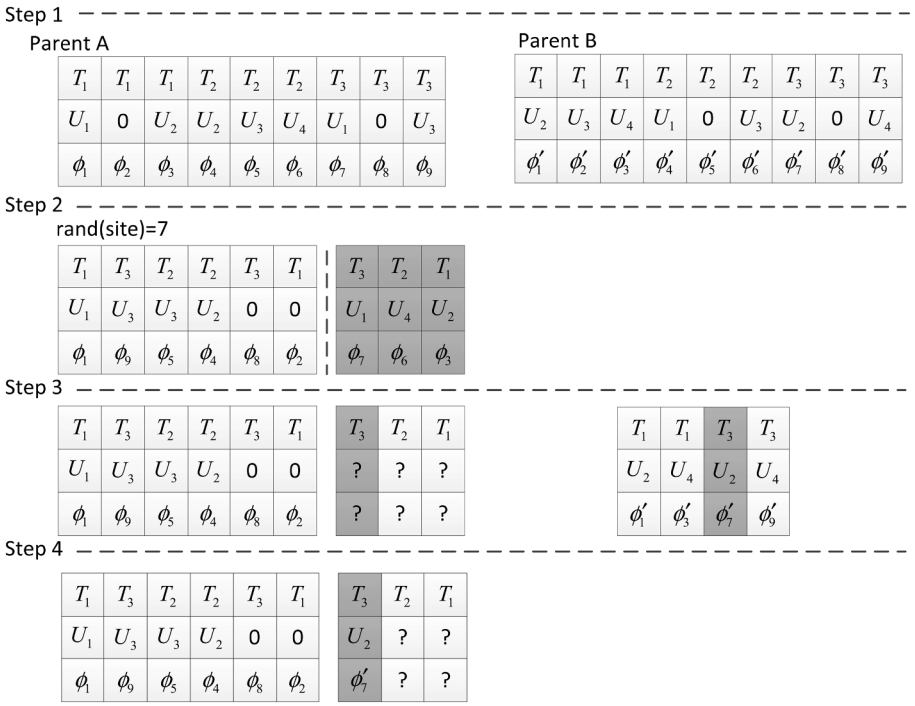


Fig. 5. An example of crossover operator.

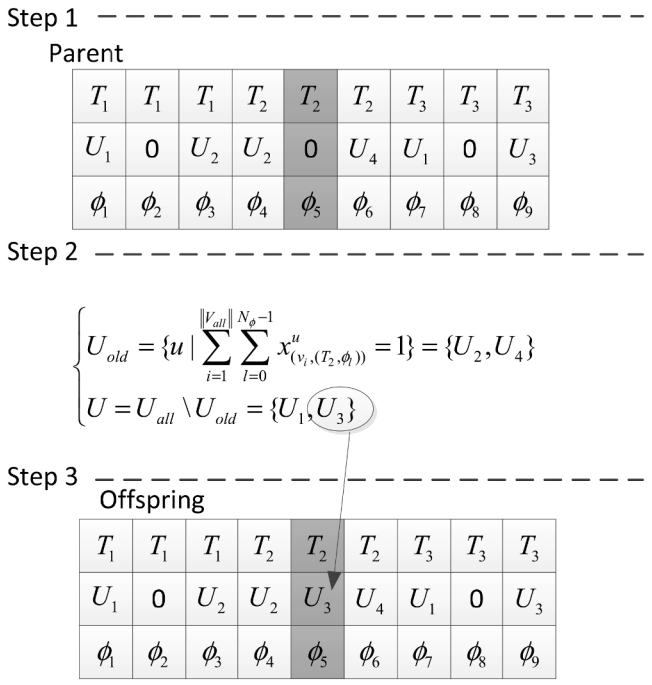


Fig. 6. An example of mutation operator: heading angle mutation

To address this, a novel chromosome unlocking method is designed to address deadlock conditions. The method involves calculating the expected utility of each target based on the situation awareness of the battlefield and subsequently determining the target execution sequence. At the initial time, the expected utility for target is determined using the initial value of the target and the positional relationship between the UAVs and the targets in the battlefield. The specific formula to compute the expected utility for the target T_j is as follows:

$$utility_j = Va_j - \beta \sum_{i=1}^{N_u} L_{ij}, \quad j \in \{1, \ldots, N_t\} \tag{19}$$

where Va_j denotes the initial value of the target T_j, L_{ij} is the Euclidean distance between the UAV U_i and target T_j, β is a user-defined coefficient, which denotes the weight of the reward and cost. The target execution sequence can be expressed as:

$$Seq_j = index\left(\boldsymbol{desend}(utility), utility_j\right), j \in \{1, \ldots, N_t\} \tag{20}$$

where Seq_j denotes the execution sequence of target T_j, $\boldsymbol{desend}(utility)$ denotes to sort the expected utility of all targets in descending order, $index(\boldsymbol{A}, a)$ denotes the position of element a in vector $\mathbf{A}$.

By determining the target execution sequence based on the situation awareness of the battlefield, the task sequence of each UAV is no longer determined by the genes sequence of the corresponding UAV-based sub-chromosome, which can also avoid the deadlock condition. To Illustrate the chromosome unlock procedure, Fig. 7 is presented as an example. The left part of Fig. 7 represents a chromosome's UAV-based sub-chromosomes. In the tradition method, the genes sequence of the UAV-based sub chromosome determines the task sequence of the corresponding UAV, which means that the task sequence of these UAVs are $A^{(1)} = [1,3,2]$,$A^{(2)} = [1,2,3]$ and $A^{(3)} = [2,1]$, respectively. The corresponding directed graph $G(T, \varepsilon(A))$ is shown in the upper right side of Fig. 7. It can be observed that the directed graph contains two directed circle, indicating that the corresponding chromosome will fall into a deadlock condition. However, with the proposed method (shown in the lower right side of Fig. 7), the target execution sequence is determined based on the situation awareness of the battlefield. Assuming the expected utility of the targets are $[40, 60, 50]$, the corresponding target execution sequence is $T_2 \rightarrow T_3 \rightarrow T_1$, leading to a directed graph $G(T, \varepsilon(A))$ without any directed circles. This signifies that the example chromosome will not fall into deadlock condition when using the proposed method. In fact, regardless of the target execution sequence, the chromosome will not encounter a deadlock condition when employing the proposed method. This is because the task sequence of each UAV is determined by the target execution sequence, and no directed circles that cause deadlock conditions will be present in the generated directed graph $G(T, \varepsilon(A))$.

4.5. UAV resource management

Considering the limited resource of UAVs (as defined in Eq. (12)) and resource requirement of targets (as specified in Eq. (13)), it is possible to allocate a UAV coalition to attack a target. The UAVs within the same coalition are required to share the resources required to attack the target. As the number of resources used by a UAV on the corresponding target cannot be obtained according to the allocation strategy from a chromosome, a resource feasibility checking process is required to check whether the chromosome satisfy the resource requirement constraint, as shown in Fig. 8.

Firstly, the target execution sequence is determined based on the

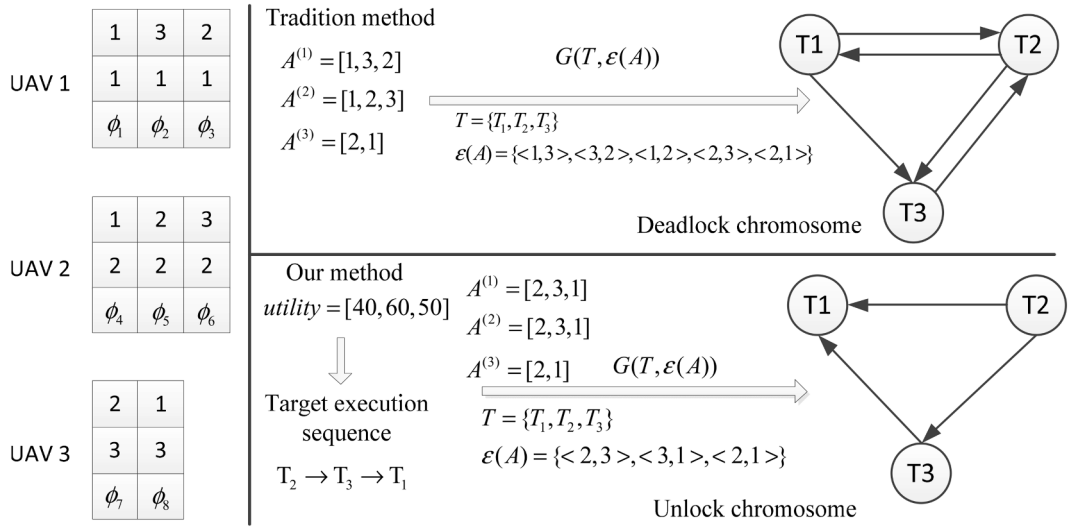


Fig. 7. Unlock the deadlock chromosome based on target execution sequence.

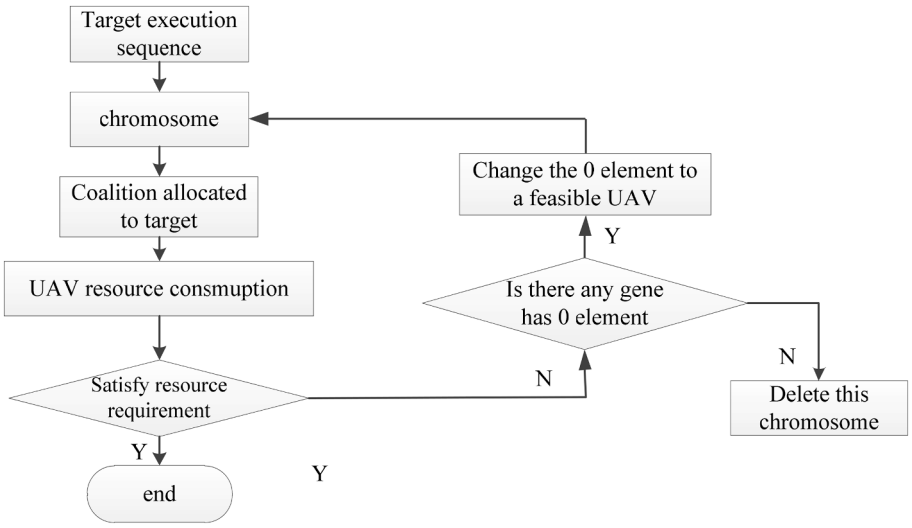


Fig. 8. Resource feasibility check of chromosome.

situation information in the battlefield. Next, the UAV coalition that performs the simultaneous attack task on each target is derived from the genes of the chromosome. Subsequently, the resource consumption of the UAVs in the coalition is calculated. If the chromosome does not meet the resource requirement constraint, a thorough check is conducted to identify whether any gene has a zero element. If any gene contains a zero element, the chromosome is modified by changing the zero element to a feasible UAV, ensuring that the resource requirement is met. On the other hand, if the chromosome does not contain any zero element but still fails to satisfy the resource requirement constraint, it is considered infeasible. In this case, the chromosome is removed, and a new chromosome is generated through the crossover operation.

In this paper, a greedy algorithm is used to calculate the resource consumption of UAVs in each coalition. For a certain UAV coalition C_j, the UAVs in this coalition are required to attack the target T_j simultaneously. The resource requirement of target T_j is represented as $R_p^{T_j}$, the resource consumption of the UAV U_k in the coalition C_j can be expressed as:

$$R_p^{U_k}(\text{cost}) = \begin{cases} 0, & if \sum_{i \in U_{b_k}} R_p^{U_i} \geq R_p^{T_j} \\ \min\left(R_p^{T_j} - \sum_{i \in U_{b_k}} R_p^{U_i}, R_p^{U_k}\right), & otherwise \end{cases} \tag{21}$$

where $R_p^{U_k}(\text{cost})$ denotes the quantity of type-p resource consumption of UAV U_k, U_{b_k} means the UAV set with more resources than U_k in the coalition C_j, which can be represented as follows:

$$\begin{cases} U_{b_k} = \left(U_m \middle| \left\{\left(U_m \in C_j\right) \& \left(\left\|R^{U_m}\right\| \geq \left\|R^{U_k}\right\|\right) \& (m \neq k)\right\}\right) \\ \left\|R^{U_m}\right\| = R_1^{U_m} + \ldots + R_n^{U_m} \end{cases} \tag{22}$$

A simple resource consumption example is illustrated in Fig. 9.

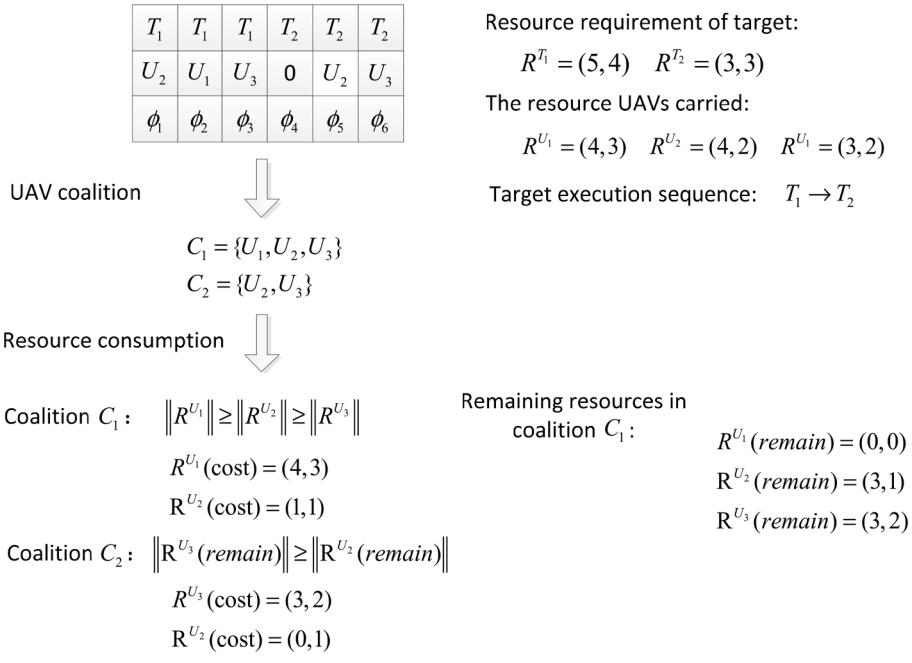


Fig. 9. An example of resource consumption of a certain chromosome (three UAVs cooperative attack two targets).

4.6. Computational complexity

To analyze the computational complexity of the modified GA, it is essential to determine the cardinality of the feasible solutions set, which corresponds to the number of feasible chromosomes generated during the algorithm's execution.

Each of the N_T targets can be allocated to a maximum number of three different UAVs. So, there are $3N_t$ genes in a chromosome. The allocation strategy of each target is represented as three genes in a chromosome. Since the target execution sequence is obtained by the situation information in the battlefield, the gene position exchange in chromosome does not affect the target execution sequence. The second row of a gene/chromosome denotes the allocated UAV, from the knowledge of probability theory, the allocation of a UAV $i \in U$ to attack a target $j_1 \in T$ and the allocation of another UAV $i_2 \in U, (i_1 \neq i_2)$ to attack the target $j_2 \in T_2, (j_1 \neq j_2)$ are independent events. So, the cardinality of feasible allocation strategies set of each target can be represented as follows:

$$N_a = (N_u + 1) \cdot N_u \cdot (N_u - 1) \tag{23}$$

To complete the computational complexity calculation of the modified GA, the heading angle at which a UAV visits a target should be considered, resulting in $(N_\varphi)^{3N_t}$ possibilities. Hence, the total cardinality of the feasible solutions set is given by:

$$N_f = [(N_u + 1)(N_u)(N_u - 1)]^{N_t} \cdot (N_\phi)^{3N_t} \tag{24}$$

Indeed, the number of feasible chromosomes can be significantly large, especially considering the size of the solution space for the integrated optimization problem. As the number of UAVs, targets, and the degree of heading angle discretization increase, the solution space expands exponentially, making it challenging to exhaustively explore all possibilities. Therefore, the modified GA is used which belongs to heuristic algorithm to seek a feasible solution of the problem.

5. Simulation results

In this section, a series of numerical simulations are conducted to showcase the enhanced performance and effectiveness of the modified GA for solving the integrated task allocation and path planning problem in multi-UAV cooperative attacking mission. In Section 5.1, an example is illustrated to show the whole process when the modified GA is used in the integrated optimization problem. In section 5.2, Monte Carlo simulation is conducted to compare the performance of the modified GA with several other methods commonly used in solving task planning problems. All of the numerical simulations are implemented on a personal PC with Intel Core i5-12400 @4.9 GHz with 16 GB of RAM.

5.1. Sample run

The investigated scenario involves four UAVs and two targets in a square region of 3200m × 3200m. The initial information of the UAVs is shown in Table 1. The two targets are located in $(1500m, 1500m), (2000m, 1000m)$ with initial value of 100 and 90, respectively. The resource requirements of targets are $R^{T_1} = (4, 2)$ and $R^{T_2} = (3, 3)$. Considering the requirement of targets, a UAV coalition maybe allocated to attack a target. The UAVs in the same coalition are required to attack the corresponding target simultaneously. The characteristics of UAVs are presented in Table 2. The parameters of the modified GA are given in Table 3.

Table 1
Initial information of UAVs.

No. UAV	Initial location (m)	Initial Heading angle (rad)	Initial resource
UAV 1	(0,0)	$\pi/6$	(2,1)
UAV 2	(1500,0)	$\pi/6$	(0,1)
UAV 3	(3000,0)	$3\pi/4$	(3,2)
UAV 4	(2000,2000)	$\pi/6$	(2,1)

At initial time, the battlefield situation is given in Fig. 10(a), where "▲" denotes the initial location of the UAVs, "☆" indicates the location of the targets. The circles and rectangles represent the obstacles and no flying zones in the battlefield environment. Then, using the modified GA, the task allocation strategy and paths for the UAV team is shown in Fig. 10(b). The computation time for this instance is 3.3 s.

It can be seen from Fig. 10(b) that, considering the resource requirement of targets, two UAV coalition are generated to attack the two targets. The UAV coalition $C_1 = [U_1, U_3]$ are required to attack the target T_1 simultaneously, coalition $C_2 = [U_2, U_3, U_4]$ needs to attack T_2 simultaneously. PH curves are generated and used as the feasible paths for these UAVs. The estimated arrival times (ETAs) for the UAVs in coalitions C_1 and C_2 are given in Fig. 11(a) and Fig. 11(b), respectively. It can be seen that, the UAVs in the two coalitions can simultaneously arrive at their allocated targets.

5.2. Monte Carlo simulations

In this section, Monte Carlo study is used to compare the modified GA with other four methods. The Particle Swarm Optimization (PSO), the Grey Wolf Optimization (GWO) and the random search method are used for comparting with the modified GA to solve the integrated optimization problem. In addition, a task allocation algorithm based on contract net protocol (CNP) is also used for comparing with the proposed method (Hooshangi et al., 2021). In the CNP-based task allocation method, the task allocation process consists three phases: ordering existing tasks, holding an auction, and implementing. In the first phase, some criteria/parameters are used to prioritize tasks. In the second phase, coordinating UAVs hold auction for the existing tasks one by one according to the ordering list, the UAV who offers the highest value for the task is the winner. In the last phase, the UAVs carry out their respective tasks. Because the CNP based algorithm can only solve the task allocation process, the simultaneous arrival algorithm used in the modified GA is used for the path planning process.

The first group of Monte Carlo simulations is to compare the performance of different methods on the mission completion time with varying number of UAVs. Two problems of different sizes in the Monte Carlo study are set. Set the "Case A" consisting of 5 targets and the "Case B" involves 10 targets. In each type of scale problems, the number of UAVs varied from 6 to 25, and 100 runs are conducted for each configuration. For each simulation run, the location of targets and UAVs are generated randomly in the battlefield environment. Each target needs two types of resource, the resource requirement of each type is an integer randomly generated between 1 and 5 (including 1and 5). The type and amount of resources carried by each UAV are similar to that of targets. The characteristics of UAV are the same as those in Section 5.1. Considering the obstacles in the battlefield environment, the number of obstacles is randomly generated between 2 and 4. In the PSO, GWO, random search and CNP-based methods, other constraints, such as the resource requirement and simultaneous arrival of targets are considered in the same way as in the proposed method. The result of average mission completion time is given in Fig. 12.

In Fig. 12(a), the horizontal axis denotes the varying number of

Table 2
Characteristics of UAVs.

Parameters	Symbols	Value
Velocity of the UAVs (m/s)	V	50
Minimum turning radius of UAV (m)	$R_{\min}$	50
Safety radius of UAV (m)	R_{safe}	100
Maximum curvature constraint (m^{-1})	$\kappa_{\max}$	0.02
Maximum control input (rad/s)	$\omega_{\max}$	1

Table 3
Parameters of GA.

Parameters	Symbols	Value
Population Size	N_p	200
Elitism Number	N_e	10
Crossover Number	N_c	180
Mutation Number	N_m	10
Generations	N_g	200

UAVs, and the vertical axis indicates the average mission completion time of 100 runs. It can be seen in Fig. 12(a) that, with the increase in the number of UAVs, the mission completion time obtained by the five algorithms will decrease. Using the random search method or the CNP based algorithm, when the number of UAV is 5, the mission completion time is similar to that of using the modified GA. However, with the increase of UAV or target number, the mission completion time decrease not significantly using the random search method and the CNP based algorithm. When using the standard PSO and the GWO, the quality of the obtained solution is not high. This is mainly because the dimension of the PSO and GWO algorithms are $N \times M$, where N and M denote the number of UAVs and targets, respectively. In addition, with the increase of UAV, the dimension of UAV increases further, so it is difficult to obtain a feasible solution with high quality in finite time. Consequently, leads the PSO and GWO algorithms to converge to local optima far from the optimal solution. The same result can also be seen in "Case B" (Fig. 12(b)). When there are only 6 UAVs in "Case B", the total resources carried by the UAVs are less than the total resources required by the target, the mission cannot be completed, so the mission completion time for 6 UAVs is omitted. When using the CNP-based task allocation method, there is a certain gap between the quality of the solution and the modified GA. Because the coupled task allocation and path planning problems consider a variety of constraints, and the CNP-based method essentially narrows the search scope of the solution, thereby obtaining suboptimal solutions. It is apparent that the performance of the modified GA is better than other four methods for the integrated optimization problem. Since all five algorithms need to use the same path planning process to obtain simultaneous arrival paths for multiple UAVs, the time consumption of these algorithms is not much different.

The second group of Monte Carlo simulation is used to compare the convergence of different methods. Here also set two-scale problems: the "Case A" with 6 UAVs, 3 targets and the "Case B" with 10 UAVs, 6 targets. 100 runs are carried out under the two scales. In each simulation run, the initial configurations of UAVs, the location and resource requirement of targets are set as randomly parameters. Fig. 13 shows the convergence performance of three methods.

In Fig. 13, the horizontal axis is the number of iterations, J_k and J_0 in the vertical axis represents the cost of the optimal individual in the k generation and initial generation, respectively. In the objective function shown in Equation (10), individuals with better performance have lower cost. The smaller the ratio J_k/J_0, the higher the quality of obtained solution.

It is evident that the modified GA outperforms the other two methods

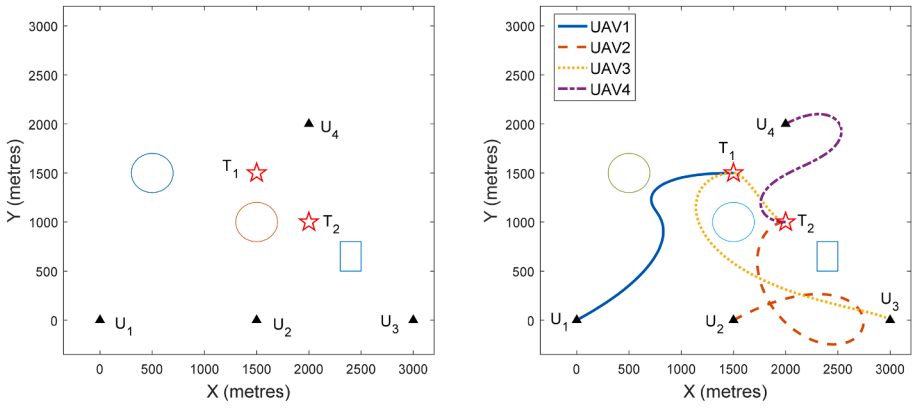


Fig. 10. (a) initial locations of targets and UAVs; (b) the task allocation and path planning results for the UAV team.

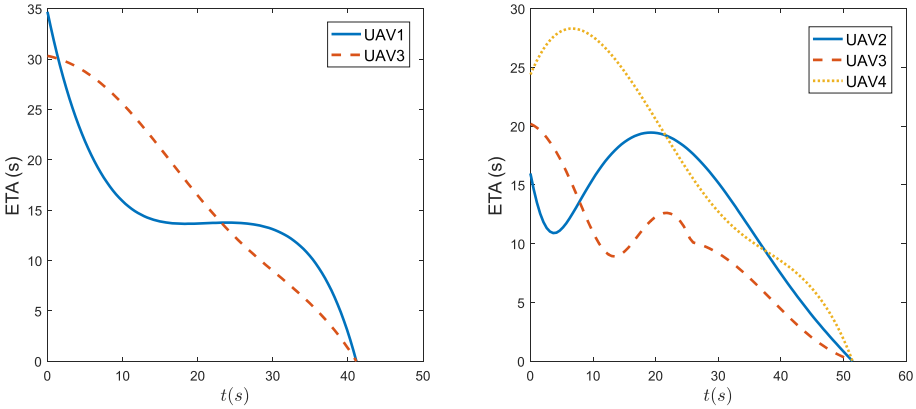


Fig. 11. (a) the ETAs for the UAVs in coalition C_1; (b) the ETAs for the UAVs in coalition C_2.

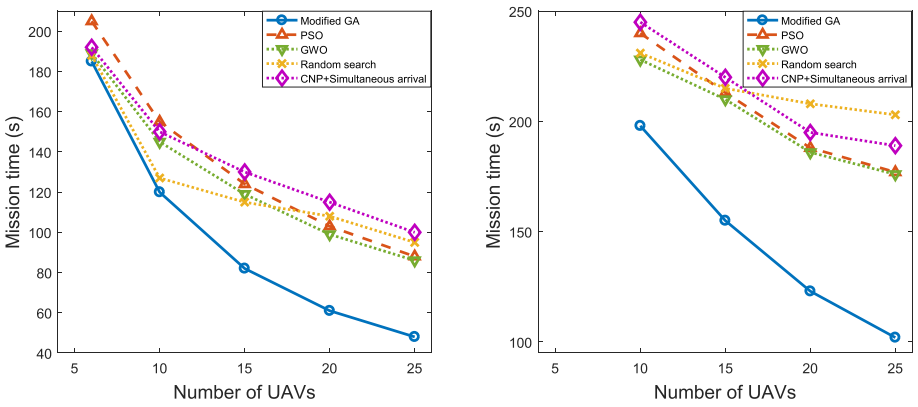


Fig. 12. The influence of five task allocation methods on mission completion time when the number of UAVs varied from 6 to 25 (a) Case A (5 targets); (b) Case B (10 targets).

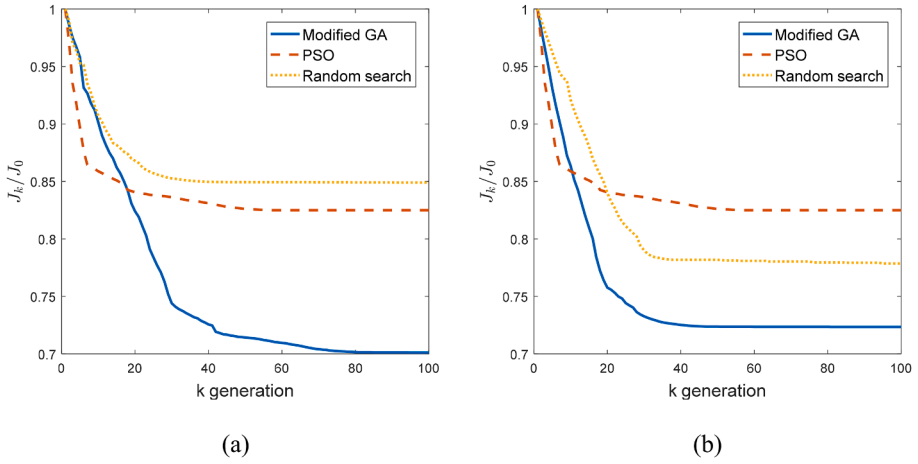


Fig. 13. The average convergence performance of three methods (a) Case A (6 UAVs, 3 targets); (b) Case B (10 UAVs, 6 targets).

for the coupled task allocation and path planning problems of varying scales. The modified GA consistently yields the highest-quality solutions, effectively minimizing the cost of multi-UAV attack missions. This improvement is attributed to the effectiveness of genetic operators, which enable the solution population of the modified GA to evolve much more efficiently compared to the pure random search method.

As a heuristic algorithm, it is generally expected to achieve performance on par with or better than the modified GA. However, in the simulations across different scales, the PSO algorithm not only underperforms the modified GA but also lags behind random search in Case B, which involves 10 UAVs and 6 targets. In this study, the coupled task allocation and path planning problems is investigated involving heterogeneous with varying mission loads allocated to targets that may require simultaneous arrival. It's important to note that the particles in PSO are moving, which can result in updates leading to infeasible positions. Various conditions, such as allocating UAVs to targets without sufficient resources, can render a position infeasible. Consequently, the abundance of infeasible candidates within the coupled task allocation and path planning problems often leads the PSO algorithm to converge towards local optima.

6. Conclusions

In this paper, a modified genetic algorithm with multi-type genes is proposed to efficiently solve the integrated task allocation and path planning problems in a multi-UAV cooperative attacking mission. By discretizing infinite possible heading angles into finite values, the coupled task allocation and path planning problems can be modelled as an integrated combinatorial optimization problem. To address resource requirement and simultaneous arrival constraints, a chromosome unlocking method is designed that deals with deadlock conditions based on the target execution sequence. This method eliminates the need to adjust the sequence of genes within a chromosome. The Monte Carlo results demonstrated that the proposed method improves the performance of multi-UAV systems in cooperative attacking missions. Although the convergence speed of the modified GA may not be the fastest, it consistently produces feasible solutions closer to optimality, enabling multi-UAV systems to complete attacking missions more efficiently.

In future research, the integrated task allocation and path planning problems in dynamic and unknown environments will be explored. The communication constraints will be taken into consideration, which will result in a time-varying communication topology for the multi-UAV network.

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CRediT authorship contribution statement

Fei Yan: Conceptualization, Methodology, Software, Writing – original draft. **Jing Chu:** Visualization, Data curation. **Jinwen Hu:** Resources, Writing – review & editing. **Xiaoping Zhu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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